Dev Patel

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SKILLS

Languages : Python, Java, SQL,C, C++, html,css

Frameworks & Libraries : YOLO, TensorFlow, PyTorch, OpenCV, scikit-learn, Hugging Face Transformers, LangChain / LlamaIndex, FastAPI / Flask

Specializations : Computer Vision, OCR Technologies, Machine Learning, Deep Learning, Big Data Analytics, Natural Language Processing (NLP), LLMs & Generative AI (RAG, Prompt Engineering, Fine-tuning)

Tools : GitHub, Docker, AWS (S3, EC2), Google Cloud (BigQuery, GCS), Azure Blob Storage, MySQL / PostgreSQL, MongoDB, Linux

PROJECTS

Real Time OCR for Partially Blind | *IoT, OCR, NLP* (Patent Pending)

- Built an IoT device helping partially blind people read written text via audio; supports multilingual translation with end-to-end latency of $\leq$**5 seconds**
- Leveraged **CRAFT (Character-Region Awareness For Text detection)** for accurate text localization and **AWS** (S3, EC2) for scalable cloud processing and inference
- Secured **$5000 in funding** (i-Create + CSIC); incubated in i-hub Gujarat; featured at Vibrant Gujarat Startup Exhibition

Smart Mark Sheet Scanner | *Python, OCR, Automation*

- Employed advanced **OCR techniques and similarity extraction** to enable faster multi-document scanning with high throughput and precision
- **Reduced workforce by 50–60%**; achieved **near 100% accuracy** via automated scanning pipeline

Crowd Counting | *Computer Vision, YOLO, Deep Learning*

- Increased confidence accuracy up to **90%** using advanced object detection techniques
- Designed for real-time monitoring and surveillance, enabling security analysis at large public events

EXPERIENCE

Research — February 2024 - April 2024

Research Intern under Prof. Ganapati Panda — IIT Bhubaneswar, India

- Developed a **CRT-based convolution algorithm** converting 1D cyclic convolutions into multi-dimensional equivalents, reducing multiplication complexity from **$O(N^2)$ to $2N-K$** — up to **$3\times$ faster** than direct convolution for ECG-length sequences
- Benchmarked the algorithm on ECG datasets via end-to-end simulation, demonstrating **measurable gains in detection accuracy** over baseline deep learning models

MVP – Missionary Visionary Products — March 2023 – Present

Co-Founder — Gujarat, India

- Engineered an assistive wearable device using a **custom ESP32 module, cloud infrastructure, Python, and ML** — capturing visual input via camera, processing it in real-time, and delivering multilingual audio output; mountable on any standard spectacle frame
- Secured grants of **$2.5k from i-Create** and **$2.5k from CSIC** for IoT accessibility project

3-Fit Internship (Start-Up) — January 2022 - May 2022

Software Engineering Intern — Gujarat, India

- Conducted research on competitive apps in similar domains to identify their strengths and weaknesses
- Integrated and worked with multiple third-party **APIs** to enhance product functionality

Teaching Assistant

- **Concordia University** – Data Structures *(During Masters)*
- **Charusat University** – AIML, UGSF (Undergraduate Student Fellowship Scheme) *(During Bachelors)*

GSA Concordia — June 2025 – June 2026

Vice-President — Montreal, QC, Canada

- Elected VP of Concordia's **Graduate Student Association (GSA)**; advocated for 5000+ graduate student interests across academic, financial, and welfare domains at the university level

EDUCATION

Concordia University — 2024 – 2026

Master of Science in Computer Science | ***GPA**: 3.3* — Montreal, Quebec, Canada

Charusat University — 2020 – 2024

B.Tech in Computer Science & Engineering | ***CGPA**: 9.5* — Changa, Gujarat, India

ACHIEVEMENTS

Winner – AZADI KA AMRIT MOHOTSAV, State-level Hackathon

Top 12 Finalist – HackX, National-level Hackathon

4th Rank – Codepie 3.0, College-level Competitive Programming Contest (200+ teams)